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Hiring Manager
AlphaSense
New York City, NY, USA

Dear Hiring Manager,

I am excited to apply for the Senior AI Engineer position within AlphaSense's Expert Research division. With over two decades of experience in software engineering and a strong focus on AI-driven systems, particularly in natural language processing (NLP) and large language models, I am eager to contribute to AlphaSense's mission of empowering professionals with AI-driven market intelligence. My extensive background in designing and deploying scalable machine learning solutions, coupled with my hands-on expertise in Python, FastAPI, and Kubernetes, aligns closely with the responsibilities of this role.

As Director of Cloud Architecture at Biophy since February 2025, I led the development of an AI-powered Retrieval-Augmented Generation (RAG) platform, architecting a modular data ingestion and scraping system using Python, Django, Pydantic, and the OpenAI API. This project leveraged Clean Architecture and SOLID principles to ensure scalability and maintainability, directly relevant to building NLP-driven features on AlphaSense's proprietary financial text datasets. Additionally, I created a custom AWS CloudWatch alerting system with Lambda, SQS, and SNS, showcasing my ability to deliver production-grade microservices, a key requirement for deploying machine learning models at AlphaSense.

At McKinsey Company from 2021 to 2025, I spearheaded the development of generative AI systems with RAG and multi-agent orchestration, accelerating mainframe migrations and data pipeline development using PySpark, Databricks, and FastAPI. My work included prompt engineering and parallel API calls with asyncio, optimizing performance for large-scale NLP tasks. I also scaled ChromaDB vector stores for a call center chatbot copilot, demonstrating my expertise in handling unstructured text data, which is central to AlphaSense's expert interview datasets. My experience deploying microservices on AWS EKS aligns with the preferred Kubernetes experience for this role.

My proficiency in the Python data and machine learning ecosystem, including PyTorch, stems from projects like the cloud cost analytics platform at Yotascale, where I architected microservices with OpenAPI and deployed them on Kubernetes with Istio. My practical, product-focused mindset, honed through delivering business-critical solutions, ensures I can prototype, build, and release features iteratively, as required at AlphaSense. Furthermore, my leadership in mentoring engineers at Biophy and McKinsey positions me to guide junior ML engineers effectively, fostering a collaborative and innovative team culture.

I am passionate about staying at the forefront of machine learning advancements, as evidenced by

my hobby projects exploring generative AI, multi-agent systems, and cutting-edge technologies like Docker and Kubernetes. I am confident that my technical expertise, leadership experience, and enthusiasm for AI-driven innovation make me a strong fit for this role. I look forward to the opportunity to contribute to AlphaSenses mission of delivering trusted insights through AI.

Thank you for considering my application. I am eager to discuss how my skills and experiences can drive value for AlphaSense. Please feel free to contact me at 856-408-4027 or via email at rongyj@hotmail.com to schedule an interview.

Sincerely,
Yongjun Rong